Matrix 2 tutorial 9

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1 Linear Differential Equations

A general n-th order linear ODE: [$\mathrm{a}_n(x)y^{(n)} + a_{n-1}(x)y^{(n-1)} + \cdots + a_0(x)y = f(x)$]

Theory

- The equation is **linear** if y and its derivatives appear only to the first power.
- **Superposition Principle:** If y_1, y_2 solve the homogeneous equation, then $c_1y_1 + c_2y_2$ also solves it.
- Solutions of homogeneous equations form a vector space.
- General solution: [$\mathrm{y} = \mathrm{y}_h + y_p$]

Problems

1. Show that if y_1, y_2 solve $y'' + y = 0$, then any linear combination also solves it.
2. Determine if $y'' + yy' = 0$ is linear.
3. Check if e^x, xe^x are linearly independent.
4. Check that $1, x, x^2, ..., x^n$ are linearly independent.

2 Constant Coefficient ODEs

[$\mathrm{a}_n y^{(n)} + \cdots + a_0 y = 0$]

Theory

- Try $y = e^{rx}$ because derivatives preserve exponential form.
- This reduces the ODE to an algebraic equation.

Characteristic Equation

$[\mathrm{a}_n r^n + \cdots + a_0 = 0]$

Cases

- Distinct roots: $[\mathrm{y} = \sum C_i e^{r_i x}]$
- Repeated roots: $[\mathrm{y} = (\mathrm{C}_1 + C_2 x + \cdots) e^{rx}]$
- Complex roots: $[\mathrm{y} = \mathrm{e}^{\alpha x}(C_1 \cos \beta x + C_2 \sin \beta x)]$

Problems

1. Solve $y'' - 4y' + 4y = 0$
2. Solve $y''' - y'' - y' + y = 0$

3 Non-Homogeneous Equations

$[\mathrm{L[y]} = \mathrm{f(x)}]$

Theory

- General solution: $[\mathrm{y} = \mathrm{y}_h + y_p]$
- Split problem into homogeneous + particular solution.

Methods

Undetermined Coefficients Works for polynomial, exponential, trig functions.

Variation of Parameters $[\mathrm{y}_p = u_1(x) y_1 + u_2(x) y_2]$

Problems

1. Solve $y'' - y = e^x$
2. Solve $y'' + 9y = \sin(3x)$

4 Non-Constant Coefficients

$[\mathrm{a}_n(x) y^{(n)} + \cdots + a_0(x) y = f(x)]$

Theory

- Exponential ansatz no longer works.
- Must use specialized techniques.

Methods

- Reduction of order
- Cauchy–Euler substitution $y = x^r$
- Power series

Problems

1. Solve $x^2y'' - xy' + y = 0$
2. Solve $xy'' + y' = 0$

5 More Problems

1. Solve $y''' + y'' - y' - y = 0$
2. Solve $y'' + 2y' + y = e^{-x}\cos x$
3. Solve $x^2y'' - 2xy' + 2y = \ln x$
4. Solve $y''' - 2y'' - y' + 2y = 0$
5. Solve $y'' + y = x\sin x$
6. Solve $y'' - 2y' + y = x^2e^x$
7. Solve $y'' + 4y = e^{2x}\cos x$
8. Solve $y'' - y = \cosh x$
9. Solve $y'' + 2y' + 2y = e^{-x}\sin x$
10. Solve $x^2y'' + xy' - y = 0$
11. Solve $x^2y'' + 3xy' + y = x^2$
12. Solve $xy'' + 2y' = x$
13. Solve $y'' + 9y = x\cos(3x)$
14. Solve $y^{(4)} - 5y'' + 4y = 0$
15. Solve $y^{(4)} + y = 0$
16. Solve $y'' - 4y = xe^{2x}$
17. Solve $y'' + 2y' + y = \ln x$

6 Solutions

Linear Differential Equations

1. If y_1, y_2 solve $y'' + y = 0$, then: [y = $c_1 y_1 + c_2 y_2 \Rightarrow y'' + y = c_1(y_1'' + y_1) + c_2(y_2'' + y_2) = 0$]

2. $y'' + yy' = 0$ is **nonlinear** due to yy'.

3. check:

4. $1, x, x^2, \ldots, x^n$ are linearly independent (polynomial basis).

Constant Coefficient ODEs

1. $y'' - 4y' + 4y = 0$
   [(r-2)$^2 = 0 \Rightarrow y = (C_1 + C_2 x)e^{2x}$]

2. $y''' - y'' - y' + y = 0$
   [(r-1)$^2(r+1) = 0 \Rightarrow y = (C_1 + C_2 x)e^x + C_3 e^{-x}$]

Non-Homogeneous Equations

1. $y'' - y = e^x$
   [y=($C_1 e^x + C_2 e^{-x}$) + $\frac{x}{2} e^x$]

2. $y'' + 9y = \sin(3x)$
   [y = $C_1 \cos 3x + C_2 \sin 3x - \frac{x}{6} \cos 3x$]

Non-Constant Coefficients

1. $x^2 y'' - xy' + y = 0$
   $y = x^r$ ...

2. $xy'' + y' = 0$
   [y = $C_1 \ln x + C_2$]

More Problems

1. $y''' + y'' - y' - y = 0$
   [y = $C_1 e^x + C_2 e^{-x} + C_3 x e^{-x}$]

2. $y'' + 2y' + y = e^{-x} \cos x$
   [y = ($C_1 + C_2 x)e^{-x} + \frac{1}{2} e^{-x} \sin x$]

3. $x^2 y'' - 2xy' + 2y = \ln x$
   [y = $C_1 x + C_2 x^2 + \frac{1}{2}(\ln x)^2$]

4. $y''' - 2y'' - y' + 2y = 0$
   $[\ \mathrm{y} = \mathrm{C}_1 e^{2x} + C_2 e^{x} + C_3 e^{-x}]$
5. $y'' + y = x \sin x$
   $[\ \mathrm{y} = \mathrm{C}_1 \cos x + C_2 \sin x - \frac{x^2}{4} \cos x]$
6. $y'' - 2y' + y = x^2 e^{x}$
   $[\ \mathrm{y} = (\mathrm{C}_1 + C_2 x) e^{x} + \frac{x^4}{12} e^{x}]$
7. $y'' + 4y = e^{2x} \cos x$
   $[\ \mathrm{y}_p = e^{2x}(A \cos x + B \sin x)]$
8. $y'' - y = \cosh x$
   $[\ \mathrm{y}_p = x(Ae^{x} + Be^{-x})]$
9. $y'' + 2y' + 2y = e^{-x} \sin x$
   $[\ \mathrm{y}_p = xe^{-x}(A \cos x + B \sin x)]$
10. $x^2 y'' + xy' - y = 0$
    $[\ \mathrm{y} = \mathrm{C}_1 x + C_2 x^{-1}]$
11. $x^2 y'' + 3xy' + y = x^2$
    $[\ \mathrm{y} = (\mathrm{C}_1 + C_2 \ln x) x^{-1} + \frac{1}{4} x^2]$
12. $xy'' + 2y' = x$
    $[\ \mathrm{y} = \mathrm{x}^2 \overline{6 + C_1 \ln x + C_2]}$
13. $y'' + 9y = x \cos(3x)$
    $[\ \mathrm{y}_p = x(A \cos 3x + B \sin 3x)]$
14. $y^{(4)} - 5y'' + 4y = 0$
    $[\ \mathrm{y} = \mathrm{C}_1 e^{x} + C_2 e^{-x} + C_3 e^{2x} + C_4 e^{-2x}]$
15. $y^{(4)} + y = 0$
    $[\ \mathrm{y} = \mathrm{e}^{x/\sqrt{2}}(C_1 \cos(x/\sqrt{2}) + C_2 \sin(x/\sqrt{2})) + e^{-x/\sqrt{2}}(C_3 \cos(x/\sqrt{2}) + C_4 \sin(x/\sqrt{2}))]$
16. $y'' - 4y = xe^{2x}$
    $[\ \mathrm{y}_p = x^2(Ax + B)e^{2x}]$
17. $y'' + 2y' + y = \ln x$
    $[\ \mathrm{y} = (\mathrm{C}_1 + C_2 x) e^{-x} + y_p(variation of parameters)]$